

Domain-Specific IT Support Chatbot Using LoRA Fine-Tuning: A Small-Scale Evaluation

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Abstract This paper reports the design and small-scale evaluation of a domain-specific IT support chatbot. A 0.5-billion-parameter instruction-tuned Qwen2.5 model was adapted with Low-Rank Adaptation (LoRA) using 94 training question answer examples. Eleven validation and eleven held-out test examples were kept separate. Three training epochs on a Google Colab T4 reduced validation loss from 1.852 to 1.655. On the held-out test set, one reviewer assigned the base model 6 of 22 possible rubric points and the adapted model 11 of 22. The result suggests more useful responses in this small sample, but does not establish production reliability. A Gradio interface permits live questions and provides example prompts. Error analysis identifies incomplete and occasionally irrelevant troubleshooting advice; therefore the system remains a prototype for low-risk first-line guidance with human escalation.

Index Terms IT support; conversational AI; Qwen2.5; LoRA; parameter-efficient fine-tuning; help desk; chatbot evaluation.

I. INTRODUCTION

I.1 Background and Motivation

First-line IT support receives short, repetitive requests about email, Wi-Fi, passwords, VPN, printers, devices, and collaboration software. A conversational interface can offer an immediate first check and help users describe an issue before a human technician takes over. Generic language models are fluent but may give advice that does not fit an organization's tools or approval process. A small domain-specific dataset provides a practical way to study whether targeted adaptation changes this behavior. The project uses an instruction-tuned open model [1] and evaluates it against its own unchanged baseline.

I.2 Research Problem

The research problem is whether a compact model trained on a limited IT help-desk dataset can answer unseen support questions more usefully than the same base model. In this setting, usefulness means relevant, safe, actionable first-line advice. It does not mean guaranteed fault diagnosis or automatic remediation. In particular, advice involving credentials, security controls, or administrator-only actions should avoid unsupported claims and should direct the user to approved support channels when needed.

I.3 Research Objectives

The project had four objectives: assemble and check an IT-specific question answer set; fine-tune an open instruction model with modest compute; compare its responses with the base model on held-out questions; and demonstrate live interaction through a browser interface. The study also documents failure cases so that a lower validation loss is not mistaken for reliable troubleshooting.

II. INDUSTRY ANALYSIS

II.1 Overview of IT Support

IT help desks handle common requests at high volume. The selected dataset spans 15 L1 support categories, including account access, email, Wi-Fi, VPN, printers, hardware, and software [2]. These categories are suitable for testing short first-response guidance. They do not represent the full variety of enterprise IT incidents, organization-specific policies, or privileged infrastructure work.

II.2 Support Structure and Challenges

A typical support workflow begins with a symptom report, checks basic connectivity or configuration, then escalates issues that require access or deeper diagnostics. Good first-line answers should prioritize safe checks, avoid pretending to inspect logs or accounts, and ask for an exact error when the symptom remains unclear. A wrong first step can waste time or expose a user to unnecessary security risk. This makes answer review as important as model fluency.

II.3 Conversational AI in IT Support

Conversational AI offers flexible language input where rigid FAQs require exact wording. Yet fluent answers can contain invented settings or omit an important diagnostic step. For IT support, evaluation should inspect whether an answer matches the symptom, gives a feasible action, and avoids harmful instructions. The present chatbot generates a response from the current question only. Its displayed chat history does not provide multi-turn memory to the model.

II.4 Potential Business Value

If validated and connected to approved knowledge, a chatbot could help staff find routine checks quickly and reduce repeated manual explanations. This project has not measured ticket deflection, time saved, user satisfaction, or business return. Those remain hypotheses for a deployment study, not outcomes of this experiment.

III. LITERATURE REVIEW

III.1 Overview

The project draws on three strands of work: small instruction-tuned language models, parameter-efficient adaptation, and broad evaluation of generated text. The cited publications provide technical background; they do not substitute for measurements on this project.

III.2 Small Instruction-Tuned Language Models

III.2.1 Qwen2.5 Technical Report

The Qwen2.5 technical report describes a family of language models and post-training methods [3]. The 0.5B Instruct checkpoint was chosen here because its size fits a Colab T4 workflow and its chat template supports concise question answer examples [1]. A smaller model is economical to experiment with, but its capacity may limit nuanced troubleshooting and domain generalization.

III.3 Parameter-Efficient Fine-Tuning

III.3.1 LoRA Study

Hu et al. introduce Low-Rank Adaptation, which freezes the original model weights and trains low-rank updates in selected layers [4]. This makes adaptation more feasible under limited memory. In this project, LoRA affected attention projection modules rather than all model parameters. Parameter efficiency reduces training cost; it does not guarantee that the adapted model learns correct support policy.

III.4 Evaluation of Language Models

III.4.1 HELM Study

The HELM study argues that language-model evaluation should use multiple scenarios and metrics instead of a single headline score [5]. This project follows that principle in a limited way by recording held-out human ratings and concrete error examples in addition to validation loss. Its small test size and single reviewer remain important weaknesses.

III.5 Related Tools and Methods

Hugging Face Transformers supplied the model and training utilities; PEFT supplied the LoRA adapter implementation [6]. Gradio Blocks supplied the browser interface [7]. Retrieval-augmented generation can ground answers in current documents [8], but it was not implemented in this chatbot. It is an appropriate direction for future organization-specific support.

III.6 Gap Addressed by This Project

A gap between broad model benchmarks and a practical help-desk prototype is the examination of actual answer quality after inexpensive domain adaptation. The present experiment is deliberately modest: few examples, one small model, and one manual reviewer. This allows transparent reporting of both gains and remaining errors.

III.7 Contribution of This Study

The contribution is a reproducible capstone pipeline that tracks data filtering, separate train/validation/test splits, LoRA settings, saved-adapter reload, before/after answer comparison, and an interactive demo. It presents the complete outcome rather than claiming enterprise readiness.

IV. METHODOLOGY

IV.1 Data Acquisition

The main source was a public synthetic English L1 IT support dataset on Hugging Face [2]. Its initial split contained 96 training, 12 validation, and 12 test examples. Four additional question answer examples were manually paraphrased from Microsoft support articles about Wi-Fi, password reset, Microsoft Store installation, and PC performance [9–12]. These four examples entered training only. The public dataset's synthetic origin avoids real ticket content but limits realism.

IV.2 Cleaning and Data Splits

Records were checked for missing fields, repeated questions, and exact question overlap across splits. Manual review rejected records containing the answer patterns `wait 15 minutes` or `re-add the work account` because these were too prescriptive without sufficient context. This removed six training, one validation, and one test example. The resulting public splits were 90/11/11; after adding four source-backed examples, the final splits were 94/11/11. Exact question overlap across splits was zero. This check does not rule out semantic similarity or answer leakage.

IV.3 Supervised Training Format

Each training record paired a user problem with an expected first-line response. The Qwen chat template encoded system and user turns as a prompt and the assistant response as the completion. Prompt tokens were masked in the language-model loss, so training targeted answer tokens. The examples were short; no vector store, retrieval index, tool calls, or private help-desk logs were used.

IV.4 LoRA Fine-Tuning

The baseline was Qwen/Qwen2.5-0.5B-Instruct [1]. A LoRA adapter used rank 8, scaling alpha 16, dropout 0.05, and `q_proj/v_proj` target modules. It trained 540,672 of 494,573,440 parameters (0.1093%). Training used three epochs, learning rate $2 \cdot 10^{-5}$, per-device batch size 2, gradient accumulation 4 (effective batch size 8), FP16, and seed 42 on a Colab T4. Validation occurred after each epoch; the best checkpoint was chosen by validation loss. The adapter and training log were saved to Google Drive, and the adapter was reloaded successfully.

IV.5 Inference and Chat Flow

At inference, the interface takes one typed question and generates up to 128 new tokens with deterministic decoding (`do_sample=False`). Gradio Blocks displays the question and answer and offers six suggested support questions. Although the UI shows previous turns, the current implementation passes only the latest question to the model. Therefore follow-up questions that depend on earlier turns may fail unless the user repeats the relevant details.

IV.6 Quality Controls and Safety

The data review removed two questionable response patterns, and final answers were reviewed manually. These controls are incomplete. The chatbot has no policy retrieval, no confidence estimate, no authentication, and no automated escalation rule. Its responses should be treated as initial guidance. In a real support setting, account lockouts, MFA problems, security incidents, and requests for privileged changes would require an approved human pathway.

IV.7 Interface and Reproducibility

The notebook stores split data, training artifacts, and response comparisons in Google Drive. The trained adapter loaded from Drive produced the same answer as the in-memory adapter on one validation question. The Gradio share link enabled a live demonstration, but it is temporary and is not a permanent hosted service. The project repository is listed in Appendix G.

V. EXPERIMENTS AND RESULTS

V.1 Evaluation Objective

The evaluation asked whether the fine-tuned model produced more useful first-line answers than the base model on questions excluded from training. Validation loss was used to monitor optimization; held-out human review assessed answer usefulness. Neither measure alone establishes factual correctness across all IT issues.

V.2 Experimental Setup

The unchanged and adapted models answered the same 11 held-out test questions under deterministic generation. One reviewer scored each answer using a three-level rubric: 0 = incorrect or misleading, 1 = partly useful, 2 = useful and correct for the question. Maximum total was 22 points per model. Scores reflect the reviewer’s judgment and are not a calibrated accuracy statistic.

V.3 Quantitative Results

Validation loss decreased from 1.852 after epoch 1 to 1.708 after epoch 2 and 1.655 after epoch 3. On the held-out test set, the base model scored 6/22 (27.3% of available rubric points) and the adapted model scored 11/22 (50.0%). The adapted model gained five points, or 22.7 percentage points of the maximum scale. No significance test was appropriate for such a small, single-reviewer sample.

Measure	Base	LoRA
Test points	6/22	11/22
Share of max	27.3%	50.0%
Validation loss		1.655

Table 1. Held-out review (11 questions).

V.4 Qualitative Observations

Several adapted answers became shorter and more relevant to first-line support, but some omitted decisive checks. For an Outlook send failure, one answer recommended checking a mobile connection and an email-server setting rather than first confirming Outlook connection status, mailbox capacity, and a simple send test. For a VPN loop after approval, the answer suggested restarting the app but did not explicitly check credentials and MFA. These examples show why lower loss and a higher aggregate score still require case-level review.

V.5 Example Cases

One test case concerning a browser extension improved from a partly useful base answer to a useful adapted answer. Another concerning device setup did not improve: both models were rated 0. A system-freeze case improved from 1 to 2. These are illustrative individual ratings from the saved test review, not a claim that all questions in those categories behave similarly.

V.6 Limitations

The evaluation set had only 11 questions across many categories; the manual rubric had one reviewer and no blind adjudication. Training data was synthetic, and the four manual examples did not cover organization-specific procedures. Exact duplicate detection did not measure paraphrase overlap. There was no live pilot, latency benchmark, security test, multi-turn test, or comparison with retrieval-based support. The temporary Gradio link and Colab runtime are unsuitable as evidence of deployment readiness.

V.7 Interpretation

The results support a narrow conclusion: in this experiment, the adapter improved the reviewer’s aggregate usefulness score and validation loss decreased across three epochs. The observed gain is promising but uncertain because the test is small and some answers remain incomplete or irrelevant. A larger evaluation with independent raters and explicit safety criteria is needed before production use.

VI. DISCUSSION

The project demonstrates that domain adaptation can change the behavior of a compact instruction model using little trainable memory. The strongest practical finding is also a caution: a fluent answer may sound supportive while missing the most useful diagnostic action. Improving the system therefore requires better source coverage and better assessment, not simply more epochs. A future version could retrieve approved, current IT knowledge, cite the source used for each answer, ask clarifying questions, and escalate uncertain cases. Those changes should be measured against the present baseline on a larger set of realistic tickets.

VII. CONCLUSION

A Qwen2.5-0.5B-Instruct model was adapted to IT support with 94 training examples and a LoRA adapter, then demonstrated in a Gradio chat interface. Across 11 held-out questions, its manual usefulness score increased from 6/22 to 11/22 relative to the unmodified model. These findings justify further investigation, not unsupervised support deployment. Future work should expand and verify data, evaluate with multiple reviewers, add approved-document grounding, test multi-turn behavior, and define escalation for security-sensitive requests.

APPENDIX

A. Application Architecture

Browser question Gradio Blocks event current-question prompt Qwen2.5 base model with LoRA adapter generated answer chat display. The base model and adapter are loaded in the Colab process; there is no external ticket-system integration.

B. Data Flow

Public synthetic dataset manual quality filter 90/11/11 clean split four source-backed training examples 94/11/11 final split tokenization three-epoch LoRA training adapter save/reload deterministic baseline/adapted inference held-out review.

C. Training Configuration

Base checkpoint: Qwen/Qwen2.5-0.5B-Instruct. Adapter: LoRA $r=8$, $\alpha=16$, $\text{dropout}=0.05$, q_proj and v_proj . Training: 3 epochs, learning rate 2×10^{-5} , batch 2, accumulation 4, FP16, seed 42. Validation loss by epoch: 1.852, 1.708, 1.655.

D. Prompt and Output Format

Input is formatted as a chat conversation with a system instruction and a user IT-support question. The assistant answer is the supervised target. Inference uses the same message format and greedy decoding. The paper does not reproduce a full prompt because the executable notebook is the authoritative version.

E. Evaluation Rubric

0: incorrect or misleading; 1: partly useful; 2: useful and correct. All 11 held-out questions were rated for both base and adapted models. Totals: 6/22 and 11/22. Ratings were made by one reviewer and should be read as preliminary human judgments.

F. User Interface

The live interface has a question box, Send button, answer history, and a side panel containing suggested questions about Outlook, VPN, Teams, printers, shared folders, and Wi-Fi. The title includes the author's name. Chat history is visual only in the current implementation.

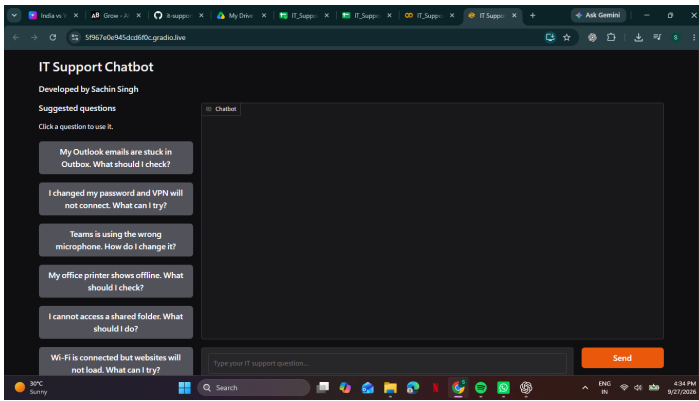


Fig. 1. Final interface with author name.

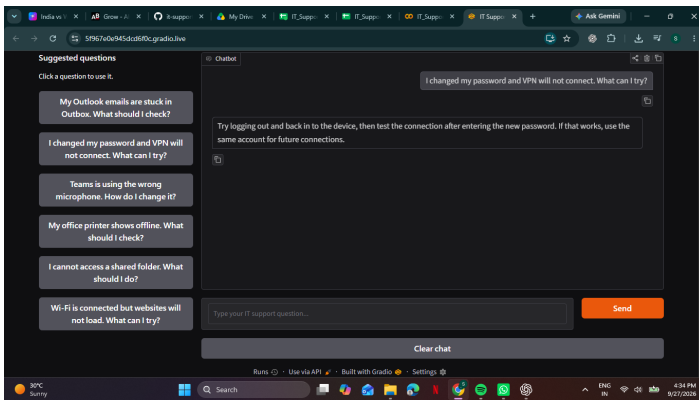


Fig. 2. Live VPN question and response.

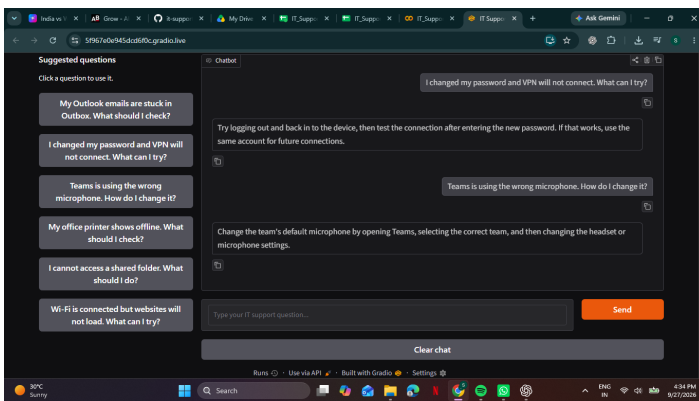


Fig. 3. Live VPN and Teams exchanges.

G. Project Materials

Project repository: <https://github.com/Sachinsinghbaghel057/it-support-chatbot>. The working notebook, cleaned JSONL splits, training log, adapter files, and scored test review are project artifacts. Availability of any shared Drive files should be checked before submission.

Live demo (temporary): <https://5f967e0e945dcd6f0c.gradio.live/>

H. Flow Diagram

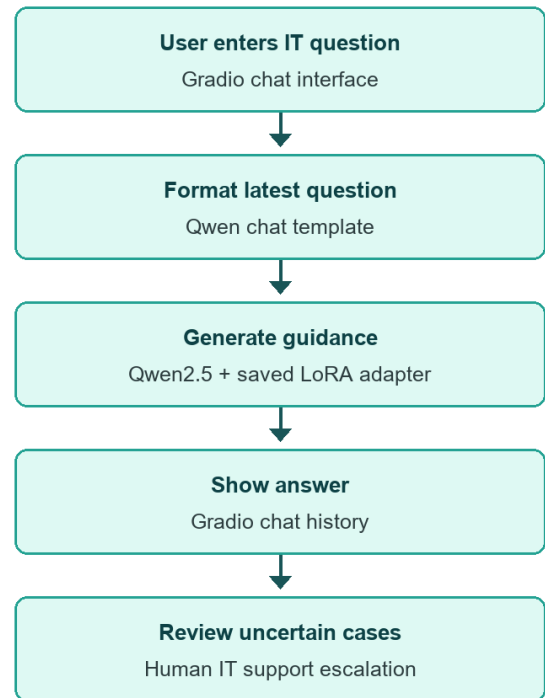


Fig. 4. Current chatbot workflow and human escalation.

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AUTHOR DETAILS

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